COM111 Tutorial 1a

1. Determine the current flowing through an element if the charge flow is given by
   a) $q(t) = (3t + 8)$ mC
   b) $q(t) = (8t^2 + 4t - 2)$ C
   c) $q(t) = (3e^{-t} - 5e^{-2t})$ nC
   d) $q(t) = 10 \sin(120\pi t)$ pC
   e) $q(t) = 20e^{-4t} \cos(50t)$ μC
2. Find the charge $q(t)$ flowing through a device if the current is:
   a) $i(t) = 3$ A, $q(0) = 1$ C
   b) $i(t) = (2t + 5)$ mA, $q(0) = 0$
   c) $i(t) = 20 \cos(10t + \pi/6)$ μA, $q(0) = 2$ μC
   d) $i(t) = 10\, e^{-30t} \sin 40t$ A, $q(0) = 0$
3. The charge flowing in a wire is plotted in Fig. 1. Sketch the corresponding current.

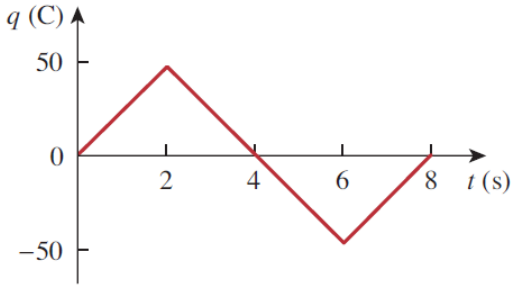


Fig. 1

4. A lightning bolt with 10 kA strikes an object for 15 μs. How much charge is deposited on the object?
5. The charge entering the positive terminal of an element is $q = 5 \sin 4\pi t$ mC while the voltage across the element (plus to minus) is $v = 3 \cos 4\pi t$ V
   a) Find the power delivered to the element at $t = 0.3$ s.
   b) Calculate the energy delivered to the element between 0 and 0.6 s.
6. The current entering the positive terminal of a device is $i(t) = 6\, e^{-2t}$ mA and the voltage across the device is $v(t) = 10$ di/dt V.
   a) Find the charge delivered to the device between $t = 0$ and $t = 2$ s.
   b) Calculate the power absorbed.
   c) Determine the energy absorbed in 3 s.
7. Fig. 2 shows the current through and the voltage across an element.
   a) Sketch the power delivered to the element for t > 0s,
   b) Find the total energy absorbed by the element for the period of 0 < t < 4s.

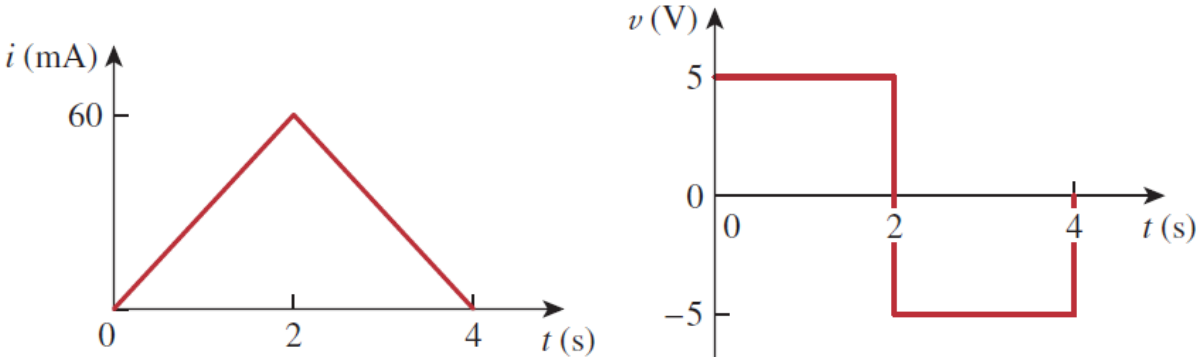


Fig. 2

8. Find the power absorbed by each of the elements in Fig. 3.

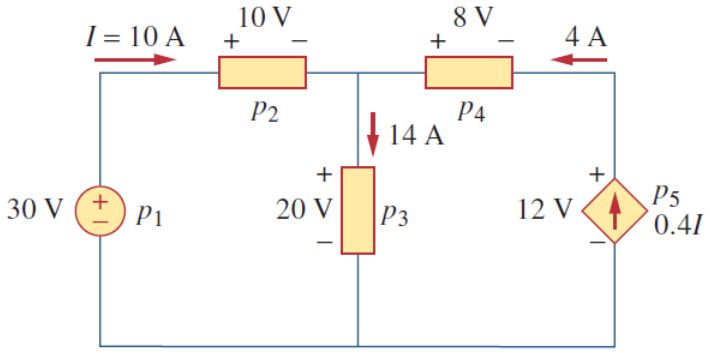


Fig. 3

9. Find V_O and the power absorbed by each element in the circuit of Fig. 4.

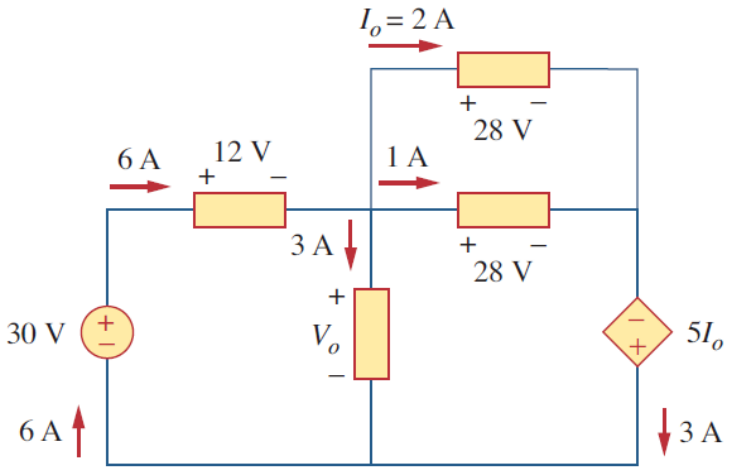


Fig. 4

10. A 1.8-kW electric heater takes 15 min to boil a quantity of water. If this is done once a day and power costs 10 cents/kWh, what is the cost of its operation for 30 days?
11. A battery may be rated in ampere-hours (Ah). A lead-acid battery is rated at 160 Ah.
    a) What is the maximum current it can supply for 40 h?
    b) How many days will it last if it is discharged at 1 mA?